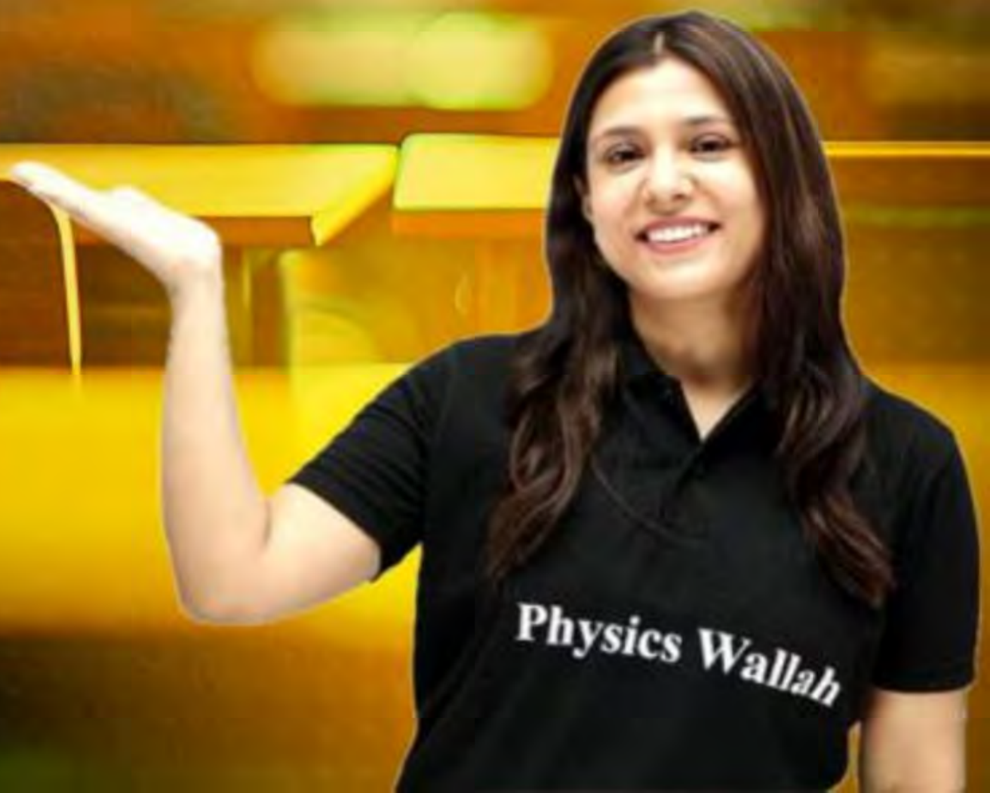
RADIANT 2026

Biology

The Flower

Lecture - 01

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Topics *to be covered*

1. Structure of a flower



Questions from Tissue

H.W

Q Name the kind of tissue -

A) Surface of skin → Epithelial Tissue - (Stratified)

B) Padding under the skin → Adipose Tissue

C) Blood vessels & lymph vessels → Squamous Epithelial Tissue.

Q. Differentiate between

A) Human Muscles [2/3 Marks]

B) Meristematic & Permanent T

C) Xylem & Phloem ✓



	Striated	Non-Striated	Cardiac
Shape	cylindrical	spindle	cylindrical & Branched
			
Nucleus			
Striation			
Function			
Location			

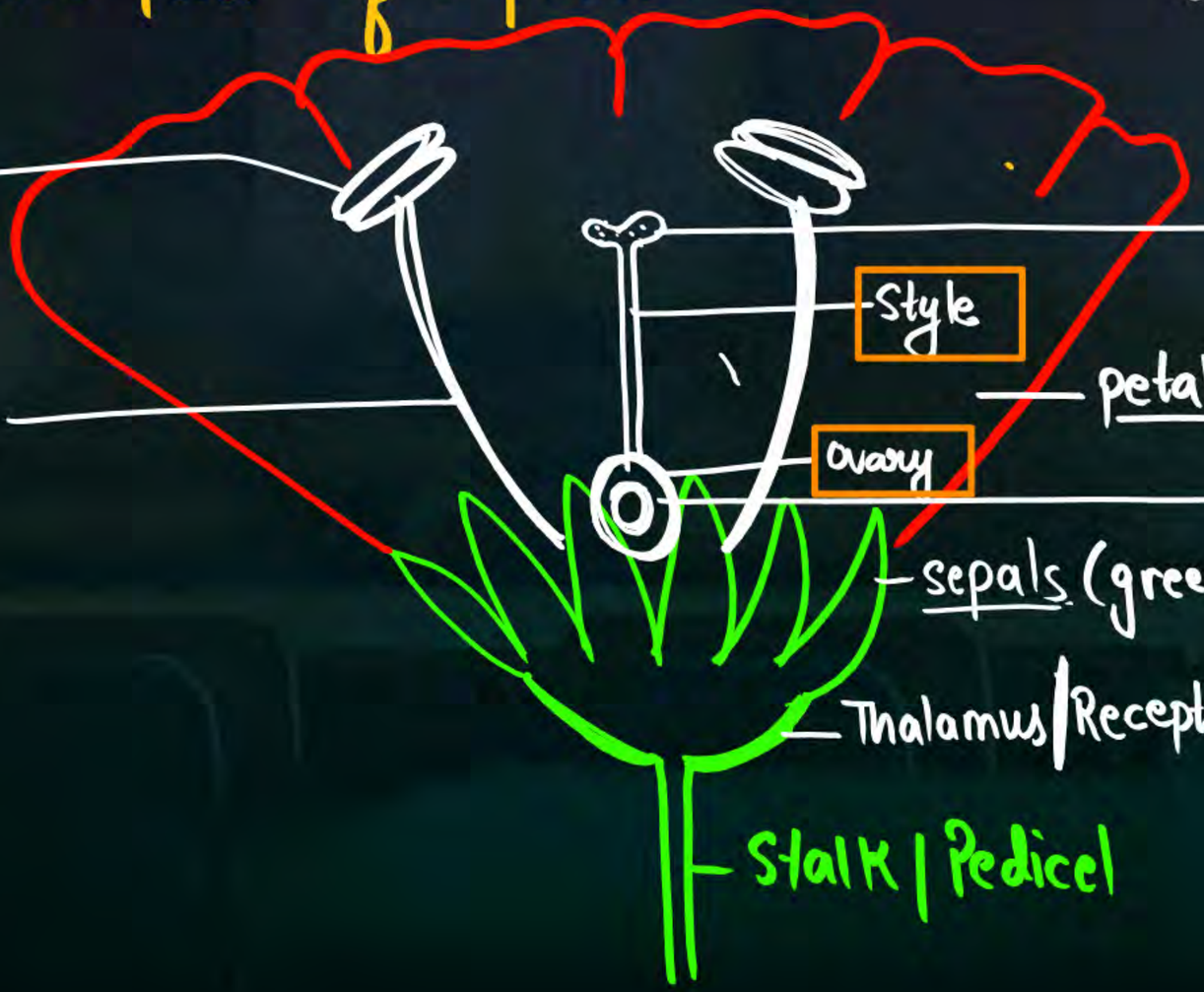
FLOWER

→ Reproductive part of a plant.

BISEXUAL

MONOECIOUS
(one house)
HERMAPHRODITE

♂ & ♀ → in same flower.



- Anthers (Bilobed)
- filament
- STAMEN [Male]
- Stigma
- Style
- Ovary
- Pistil / Carpel [female]
- petals (colourful)
- ovule
- sepals (green)
- Thalamus / Receptacle
- Stalk / Pedicel

Group.

Sepals	Calyx.
Petals	Corolla
Anther	Androecium(♂)
Pistil	Gynoecium(♀)

[Sepals + Petals]
Tepals

Perianth

Sepaloid Perianth
Petaloid Perianth.

UNISEXUAL / DIOECIOUS

either ♂ or either ♀ in one flower.

Pistillate Flower

♀

- A flower that lacks stamens and only has pistils are known as pistillate flowers. Chrysanthemum, cucumber, eggplant, squash



Staminate Flower

- Flowers that lack pistils and only have stamens are called staminate flowers. Chrysanthemum, cucumber, eggplant, squash.

Unisexual

♂ ♀



Structure of a Bisexual Flower



The floral parts are borne on the thalamus in four whorls.

First whorl – green sepals (collectively called calyx).

The sepals may be free (polysepalous) of fused (gamosepalous).



Polysepalous rose





Gamosepalous hibiscus



Gul mohar (colourful sepals)

Structure of a Bisexual Flower

- Second whorl – large brightly-coloured petals (collectively called corolla).



Double whorl
(poppy)



Spiral whorl
(water lily)



Gamopetalous
(morning glory)

Structure of a Bisexual Flower

- Third whorl (male parts) (♂ = symbol for male) – long thread-like structures somewhat projecting out each usually ending in a bilobed tip.

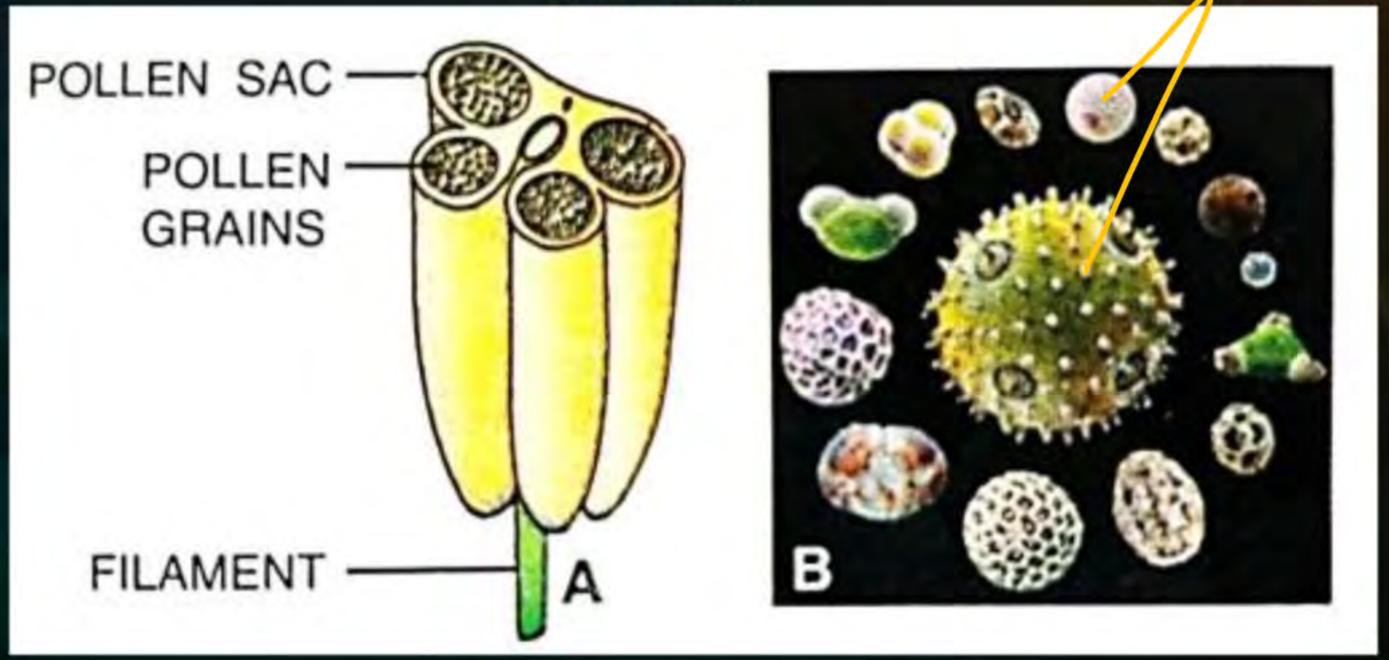
(Filament)

(Anther)

pollen grains



Polyandrous (petunia)





1 lobe ⇒ 2 pollen sacs.
1 anther ⇒ 4 pollen sacs.

Joined Stamen

♂ (anther + filament)

- Stamens are Joined in Groups

Adelphous → group of brother



Monoadelphous
(China rose)



Diadelphous
(pea)



Polyadelphous
(*bombax*)

Structure of a Bisexual Flower

- Fourth whorl– (female parts) (♀ = symbol for female) – centrally located pistil that may be formed of a single female unit called carpel. Each carpel consists of a basal ovary, a middle style and an uppermost stigma.
- Stigma– it may be divided into two or more lobes and sometimes feathery.
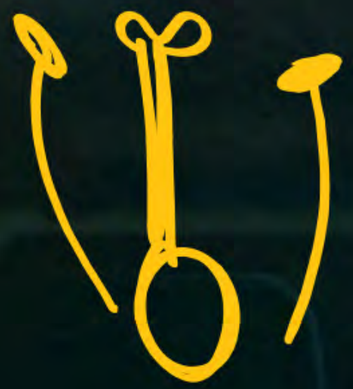






Complete and Incomplete Flowers

- A complete (or perfect) flower is one which contains all the four floral whorls. If one or more sets of floral whorls are missing, the flower is called incomplete (or imperfect).

Neuter flower

Essential and Nonessential Whorls

Which are important Reproductive parts



Gynoecium

Androecium

- Sepals
- Petals

Question

The essential whorls of a flower are the

A. Calyx and corolla

B. Stamen and ovary

C. Calyx and epicalyx

D. Androecium and gynoecium

Tepals (Perianth)

- When petals and sepals cannot be differentiated from each other, then it is called tepals collectively called Perianth.

- Green-Sepaloid perianth
- Non-Green- Petaloid perianth



Tulip

Bracts

- A modified leaf or scale, typically small, with a flower or flower cluster in its axil.
- Bracts are sometimes larger and more brightly coloured than the true flower.



Bougainvillea

Sexuality in Flowers

- BISEXUAL / HERMAPHRODITE/MONOECIOUS - A flower which contains both stamens and carpels e.g. hibiscus, rose lily and tomato, maize, cucumber, pumpkin

- UNISEXUAL / INCOMPLETE/DIOECIOUS - A flower which has only one of these parts, that is only the stamens or only the carpels, e.g. papaya and palm.

Question

Stigma may be simple or divided into two or more lobes.

A True

B False

Question

When a flower has both androecium and gynoecium

→ ♀ & ♂

A Bisexual

B Monoecious [एक घर 

C Dioecious

D Both A & B

Question

Which of the following shows diadelphous stamens?

A multi groups of carpel.

B 2 groups of Carpel. ✗

C 2 groups of Androecium.

D multigroups of stamen.
↳ poly (free).

Chrysanthemum is an example of _.

A) Unisexual

♂ ♀

B) Hermaphrodite

C) Bisexual.

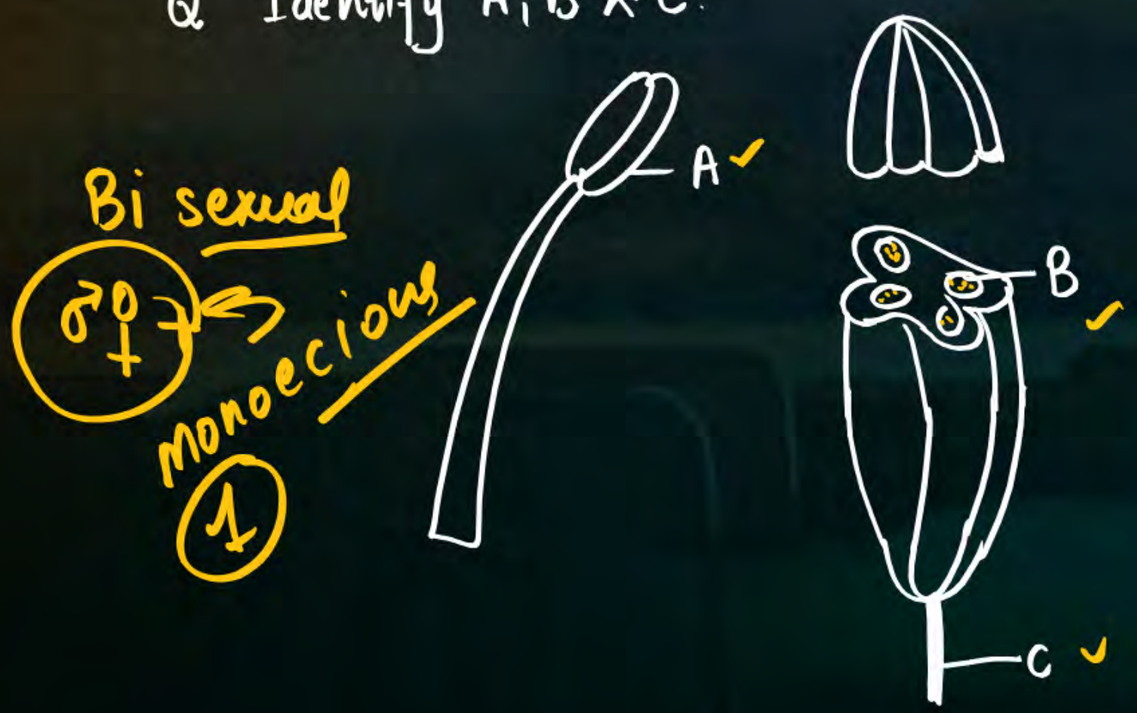
D) None of the above.

Homework from Radiant 2026 Module

Q. Differentiate between Monoadelphous, Diadelphous & Polyadelphous.

Q. What are Bracts ? State their function.

Q Identify A, B & C.




Thank You